

Table 4: Estimates of the union premium

|                                  | Estimate | Standard error | N     |
|----------------------------------|----------|----------------|-------|
| $\hat{\beta}_{fe}$               | 0.107    | 0.030          | 4,360 |
| $\hat{\beta}_{fd}$               | 0.060    | 0.032          | 3,815 |
| DID <sub>M</sub>                 | 0.041    | 0.034          | 3,815 |
| DID <sub>M</sub> <sup>pl</sup>   | 0.094    | 0.038          | 3,101 |
| DID <sub>M</sub> <sup>pl,2</sup> | -0.041   | 0.030          | 2,458 |
| DID <sub>M</sub> <sup>pl,3</sup> | -0.004   | 0.033          | 1,881 |

*Notes.* This table reports estimates of the effect of the union premium, as well as placebo estimators of the common trends assumption. Estimators are computed using the data of Vella and Verbeek (1998). Standard errors are clustered at the worker level.

## 7 Conclusion

Almost 20% of empirical articles published in the AER between 2010 and 2012 use regressions with groups and period fixed effects to estimate treatment effects. In this paper, we show that under a common trends assumption, those regressions estimate weighted sums of the treatment effect in each group and period. The weights may be negative: in one application, we find that almost 50% of the weights are negative. The negative weights are an issue when the treatment effect is heterogeneous, between groups or over time. Then, one could have that the treatment's coefficient in those regressions is negative while the treatment effect is positive in every group and time period. We therefore propose a new estimator to address this problem. This estimator estimates the treatment effect in the groups that switch treatment, at the time when they switch. It does not rely on any treatment effect homogeneity condition. It is computed by the `fuzzydid` and `did_multiplgt` Stata packages. In the two applications we revisit, this estimator is significantly and economically different from the two-way fixed effects estimators.

## References

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## A Proofs

### One useful lemma

Our results rely on the following lemma.

**Lemma 1** *If Assumptions 1-5 hold, for all  $(g, g', t, t') \in \{1, \dots, G\}^2 \times \{1, \dots, T\}^2$ ,*

$$\begin{aligned} & E(Y_{g,t}|\mathbf{D}) - E(Y_{g,t'}|\mathbf{D}) - (E(Y_{g',t}|\mathbf{D}) - E(Y_{g',t'}|\mathbf{D})) \\ &= D_{g,t}E(\Delta_{g,t}|\mathbf{D}) - D_{g,t'}E(\Delta_{g,t'}|\mathbf{D}) - (D_{g',t}E(\Delta_{g',t}|\mathbf{D}) - D_{g',t'}E(\Delta_{g',t'}|\mathbf{D})). \end{aligned}$$

*Proof of Lemma 1*

For all  $(g, t) \in \{1, \dots, G\} \times \{1, \dots, T\}$ ,

$$\begin{aligned} E(Y_{g,t}|\mathbf{D}) &= E\left(\frac{1}{N_{g,t}} \sum_{i=1}^{N_{g,t}} Y_{i,g,t} \middle| \mathbf{D}\right) \\ &= E\left(\frac{1}{N_{g,t}} \sum_{i=1}^{N_{g,t}} (Y_{i,g,t}(0) + D_{i,g,t}(Y_{i,g,t}(1) - Y_{i,g,t}(0))) \middle| \mathbf{D}\right) \\ &= E(Y_{g,t}(0)|\mathbf{D}) + D_{g,t}E(\Delta_{g,t}|\mathbf{D}) \\ &= E(Y_{g,t}(0)|\mathbf{D}_g) + D_{g,t}E(\Delta_{g,t}|\mathbf{D}), \end{aligned}$$

where the third equality follows from Assumption 2, and the fourth from Assumption 3. Therefore,

$$\begin{aligned} & E(Y_{g,t}|\mathbf{D}) - E(Y_{g,t'}|\mathbf{D}) - (E(Y_{g',t}|\mathbf{D}) - E(Y_{g',t'}|\mathbf{D})) \\ &= E(Y_{g,t}(0) - Y_{g,t'}(0)|\mathbf{D}_g) - E(Y_{g',t}(0) - Y_{g',t'}(0)|\mathbf{D}_{g'}) \\ &+ D_{g,t}E(\Delta_{g,t}|\mathbf{D}) - D_{g,t'}E(\Delta_{g,t'}|\mathbf{D}) - (D_{g',t}E(\Delta_{g',t}|\mathbf{D}) - D_{g',t'}E(\Delta_{g',t'}|\mathbf{D})) \\ &= E(Y_{g,t}(0) - Y_{g,t'}(0)) - E(Y_{g',t}(0) - Y_{g',t'}(0)) \\ &+ D_{g,t}E(\Delta_{g,t}|\mathbf{D}) - D_{g,t'}E(\Delta_{g,t'}|\mathbf{D}) - (D_{g',t}E(\Delta_{g',t}|\mathbf{D}) - D_{g',t'}E(\Delta_{g',t'}|\mathbf{D})) \\ &= D_{g,t}E(\Delta_{g,t}|\mathbf{D}) - D_{g,t'}E(\Delta_{g,t'}|\mathbf{D}) - (D_{g',t}E(\Delta_{g',t}|\mathbf{D}) - D_{g',t'}E(\Delta_{g',t'}|\mathbf{D})), \end{aligned}$$

where the first equality follows from Assumption 3, the second from the linearity of the conditional expectation and Assumption 4, and the third from Assumption 5.

### Proof of Theorem 1

It follows from the Frisch-Waugh theorem and the definition of  $\varepsilon_{g,t}$  that

$$E(\hat{\beta}_{fe}|\mathbf{D}) = \frac{\sum_{g,t} N_{g,t} \varepsilon_{g,t} E(Y_{g,t}|\mathbf{D})}{\sum_{g,t} N_{g,t} \varepsilon_{g,t} D_{g,t}}. \quad (4)$$